

App Product Analytics Dataset - SQL & Power BI

Question Set + Data Dictionary

1) Dataset Summary

This dataset represents a subscription-based mobile/web app with users, sessions, events, subscriptions, payments, refunds, support tickets, and experiments. It is designed for Product Analytics practice (funnels, engagement, retention, monetization).

2) Table List (Data Dictionary - Overview)

Table	What it contains	Primary Key	Key Relationships
users	User profile + acquisition + device	user_id	1:* sessions, events, subscriptions, payments, refunds, support tickets
plans	Subscription plan catalog	plan_id	1:* subscriptions
subscriptions	Subscription lifecycle per user	subscription_id	*:* users via user_id; *: plans via plan_id; 1:* payments
sessions	App sessions	session_id	*:* users via user_id; 1:* events
events	Event log inside sessions	event_id	*:* sessions via session_id; *: users via user_id
feature_usage	Daily feature usage counts	usage_id	*:* users via user_id
payments	Payment transactions for subscriptions	payment_id	*:* subscriptions via subscription_id; *: users via user_id
refunds	Refunds linked to payments	refund_id	*:* payments via payment_id; *: users via user_id
support_tickets	Customer support interactions	ticket_id	*:* users via user_id
experiments	A/B experiments metadata	experiment_id	1:* experiment_assignments
experiment_assignments	User assigned to variants	assignment_id	*:* experiments via experiment_id; *: users via user_id
data_dictionary	Column-level documentation	-	Reference only

3) SQL Question Series (20 Questions)

- 1) Total users, active users (`is_active = true`), and inactive users count.
- 2) New users trend: signups by month and top acquisition channels per month.
- 3) Top countries and cities by user count (validate geography distribution).
- 4) Sessions trend: sessions per day and average sessions per user.
- 5) Average session duration (seconds) by `device_type` and `app_version`.
- 6) DAU: daily active users using events table (distinct `user_id` per day).
- 7) MAU: monthly active users and stickiness (DAU/MAU).
- 8) Event distribution: top `event_type` and their share (%) of total events.
- 9) Funnel conversion: `app_open` → `content_view` → `feature_use` → `purchase_success` (user-level).
- 10) Feature adoption: top 10 features by total `usage_count` and unique users.
- 11) Feature retention: users who used any feature in week 1 and returned in week 2.
- 12) Subscription conversion rate: % users who started a trial or clicked subscribe and later paid successfully.
- 13) Revenue trend: total successful payment amount by month and by plan tier.
- 14) ARPPU: average revenue per paying user (success payments only).
- 15) Payment failure rate by `payment_method` and its trend over time.
- 16) Refund rate: % successful payments that resulted in a refund and total refunded amount.
- 17) Support load: tickets per 1,000 users and resolution rate (resolved/closed).
- 18) SLA performance: % tickets resolved within SLA hours (approx using created/resolved dates).
- 19) Churn proxy: users with canceled/expired subscription and no sessions in last 30 days.
- 20) Experiment impact: compare conversion (`purchase_success`) between Variant A vs B for each experiment.

4) Power BI Dashboard Tasks / Questions (20 Tasks)

- 1) Build relationships (star schema): users as dimension; sessions/events/subscriptions/payments/refunds as facts.
- 2) Create KPI cards: Total Users, Active Users, DAU, MAU, Sessions, Revenue, Paying Users, Refund Rate.
- 3) Create a Date table and mark it as Date Table (required for time intelligence).
- 4) Engagement page: DAU trend, sessions trend, average session duration by device.
- 5) Acquisition page: signups by channel, country, device, and conversion to paid users.
- 6) Funnel page: app_open → content_view → feature_use → purchase_success conversion rates.
- 7) Feature usage page: top features by usage_count and unique users; feature adoption over time.
- 8) Monetization page: revenue by plan tier and billing cycle; ARPPU and revenue trend.
- 9) Payments page: success vs failed payments by method; failure rate trend.
- 10) Refunds page: refunded amount trend and refund reasons distribution.
- 11) Subscription lifecycle: active vs canceled vs expired over time; churn proxy metrics.
- 12) Cohort analysis: signup month cohorts vs monthly activity (retention heatmap).
- 13) User segmentation: revenue and engagement by age_group, gender, device_type, country.
- 14) Support page: ticket volume by category/priority and resolution rate.
- 15) SLA page: % tickets resolved within SLA by category and priority.
- 16) Experiment analysis: conversion rate by variant A vs B (per experiment).
- 17) Drill-through: user profile page (sessions, events, payments, tickets).
- 18) Drill-through: plan detail page (subscriptions, revenue, churn).
- 19) Add tooltips and bookmarks (executive summary view vs detailed view).
- 20) Publish-ready dashboard: consistent formatting, slicer panel, and performance optimization.